HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Architected and implemented **Python** data processing pipelines on **AWS Lambda** and **S3** to ingest and transform ~3,500 events per second during peak hours, replacing a less efficient batch job.
- Developed **Java Spring Boot** microservices to expose REST APIs for cross-service data exchange, incorporating request validation and circuit-breaker patterns for enhanced resilience.
- Managed production support for enterprise applications using **Splunk** and **Dynatrace**, responding to P1/P2 incidents and ensuring adherence to strict SLAs for a critical financial services dashboard.
- Designed and executed reliability testing protocols for novel semiconductor devices, including electrochemical biosensors and magnetoresistive sensors, to characterize lifetime performance and identify potential failure modes.
- Developed statistical lifetime models using **Python** and **JMP** to evaluate technology robustness and predict device longevity under various stress conditions.
- Collaborated with failure analysis and process development teams to systematically investigate and determine the root cause of device failures, feeding insights back into design and manufacturing improvements.
- Containerized applications using **Docker** and deployed them onto **AWS EC2** instances, configuring basic resource limits and monitoring through **CloudWatch** for predictable operational performance.

SKILLS

Languages – Python, Java, SQL, R

Frameworks & Libraries – Spring Boot, pandas, NumPy, scikit-learn, JMP

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Databases – PostgreSQL, MySQL, Oracle

Tools & Platforms – Splunk, Dynatrace, Jira, Confluence

Data Analytics & BI – Power BI, Tableau, DAX, Power Query, Data Modelling

Engineering Disciplines – Reliability Engineering, Device Testing, Statistical Modelling, Root Cause Analysis

PROJECTS

Reliability Characterization of Novel Sensors

- Developed bespoke stress tests and lifetime acceleration techniques to de-risk novel electrochemical and semiconductor devices, moving beyond industry-standard testing protocols.
- Designed and executed comprehensive device reliability testing campaigns, documenting results and identifying key parameters affecting device longevity for early-stage technology development.
- Collaborated with cross-functional engineering teams, including materials science and device physics experts, to interpret test data and inform design iterations.

Automated Data Pipeline for System Monitoring

- Engineered **Python** data pipelines to automate the ingestion and processing of system logs and performance metrics from diverse sources, feeding into an operational monitoring dashboard.
- Implemented data validation and transformation steps using **pandas** and **SQL**, ensuring data quality and consistency for downstream analysis and alerting functionalities.
- Scheduled and managed pipeline execution using **AWS Lambda** and **CloudWatch** events, establishing automated workflows that reduced manual data collation effort by approximately 15 hours per week.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)